

VANCO AI Assessment

Solution Document

Grocery Forecasting · ASL Detection · Physics RAG

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Assessment Overview

Use Case	Type	Model	Key Result
1 · Grocery Forecasting	Tabular ML / Time-Series	LightGBM (1953 trees, 64 features)	Kaggle RMSLE 0.422
2 · ASL Detection	Computer Vision	YOLOv8n (6 MB, 29 classes)	mAP@0.50=90.6%, 45 FPS
3 · Physics RAG	NLP / RAG	BM25+ChromaDB+KG+LLaMA-3	~2s latency, grounded

Repository: <https://github.com/Mihir2603/vanco-ai-assessment>

Use Case 1 — Grocery Sales Forecasting

Problem: Forecast 16-day daily unit sales for ~1,800 store×product-family combinations at Corporación Favorita supermarkets (Kaggle Store Sales competition).

Pipeline

Raw CSVs → Merge 6 tables → Feature engineering (8 groups, 64 features) → Walk-forward CV → LightGBM → submission.csv

Feature Engineering (64 features)

Feature Group	Features
Calendar	year, month, day-of-week, is_weekend, is_payday (15th + month-end)
Lag	lag_16 / 21 / 28 / 35 / 42 / 49 / 56 per store × family
Rolling stats	mean / std / max / min over 7, 14, 28 days
Promotion	onpromotion flag, promo_count_7/14, days_since_promo
Oil price	level, oil_lag1/7, daily change, 7-day moving average
Holiday	national/regional/local flags, pre/post ±1–3 days, holiday streak
Transactions	transactions_lag per store (deduplicated), roll7/14
Encodings	store_mean_log, family_mean_log, store_family_mean_log

Critical Design Decision — Safe Lags (≥ 16 days)

The forecast horizon is 16 days. All lag features start at **lag_16** to prevent NaN at test time. Shorter lags (lag_1, lag_7) caused catastrophic failures (NaN → model predicted ~27 instead of ~2600 → Kaggle 2.54). Fixing to safe lags improved the score to **0.42234**.

Validation Strategy

Walk-forward backtesting — 5 folds, expanding training window, 16-day gap, 16-day validation. No random splits. Fold 4 covers final weeks of 2017 to mirror test distribution and prevent temporal leakage.

Model: LightGBM

- **Why LightGBM:** 3M training rows → trains in ~20 min. Neural models (TFT, N-BEATS) need >2h without guaranteed improvement.
- **Log1p target:** Training on $\log_{1p}(\text{sales})$ + RMSE is equivalent to RMSLE minimisation; back-transform with $\exp m1$.
- **1953 trees:** Trained on full data using optimal tree count from CV.

Results

0.42234	0.3897	0.6558	–40.6%
Kaggle Public RMSLE	Holdout RMSLE	Baseline (Naïve)	Improvement

Metric	Value
Baseline RMSLE (Seasonal Naïve)	0.6558

Metric	Value
LightGBM Holdout RMSLE	0.3897
Improvement over baseline	40.6%
Kaggle Public RMSLE (submitted)	0.42234

Error Analysis

- **Worst stores:** Stores 19, 20, 22 (RMSLE ~0.47–0.49) — high-volume stores in volatile regions.
- **Worst families:** SCHOOL AND OFFICE SUPPLIES (0.662), POULTRY (0.619), LAWN AND GARDEN (0.584).
- **Top features:** rolling_mean_7 > lag_21 > rolling_mean_14 — recent rolling statistics dominate.

Limitations & Next Steps

Limitation	Proposed Fix	Expected Gain
No hyperparameter tuning	Optuna Bayesian search	~0.01–0.02 RMSLE
Single recursive model	Direct multi-step (one model/day)	Better tail accuracy
No neural models	TFT or N-BEATS	Long-range seasonality
Homogeneous per store×family	Cluster micro-models by store type	Reduces heterogeneity

Use Case 2 — ASL Alphabet Detection

Problem: Real-time detection and classification of 29 ASL hand signs (A–Z + del + nothing + space) using YOLOv8 object detection, with live webcam demo.

Pipeline

Kaggle images → MediaPipe auto-annotation (21 landmarks → bbox) → YOLO format dataset → YOLOv8n training (30 epochs) → Live webcam demo (45 FPS, CPU)

Dataset

Property	Value
Source	Kaggle: grassknotted/asl-alphabet
Original size	87,000 images (3,000/class × 29 classes)
Subsampled	300/class × 29 = 8,700 images
Train / Val / Test	70% / 20% / 10% (stratified)
Annotation format	YOLO .txt — cx cy w h (normalised)
Bounding box method	MediaPipe 21-landmark hand bbox + 20% padding

Why Auto-Annotation with MediaPipe?

The Kaggle dataset is classification-only (no bounding boxes). MediaPipe detects 21 hand landmarks; we compute a padded bounding box automatically. Reproducible and consistent — no manual labelling for 8,700 images.

Model & Design Decisions

Decision	Choice	Rejected	Reason
Architecture	YOLOv8n	Faster R-CNN	Single-stage: 3–5× faster, live FPS critical
Model size	Nano (6 MB)	Small/Medium	CPU-only deployment; nano gives 45 FPS
Pre-screening	MediaPipe	Always-on YOLO	Skip YOLO on empty frames → ~60% CPU load
Augmentation	HSV+rotation+mosaic	None	Improves generalisation across lighting

Results

90.6%	90.5%	89.5%	90.0%	45.4
mAP@0.50	Precision	Recall	F1	FPS (CPU)

Class	AP@0.50	Notes
B, D, F, H, O, P, S, T	99.5%	Highly distinctive hand shapes
C	79.5%	Curved shape confused with O at distance
W	59.5%	Similar to V; 3-finger spread ambiguous

Class	AP@0.50	Notes
space	39.5%	Open palm similar to B; context-dependent

Live Demo

```
cd asl_detector && python demo.py --weights models/asl_yolov8_best.pt --conf 0.45
```

Limitations & Next Steps

- **Signer-independent evaluation:** Kaggle images have plain backgrounds; real-world users may see lower accuracy.
- **J and Z require motion:** Static frame detection is insufficient for these letters.
- **Background diversity:** Retrain with varied backgrounds for production robustness.
- **Custom data:** scripts/collect_data.py — retrain on personal webcam data for personalisation.

Use Case 3 — NCERT Physics RAG

Problem: Production-quality RAG system for NCERT Class 12 Physics Q&A; using three complementary retrieval strategies fused with Reciprocal Rank Fusion (RRF), with live web chat demo.

Architecture

```
User Question | +---> Semantic Search (ChromaDB + MiniLM-L6-v2) ---> top-k chunks +---> Keyword Search (BM25 Okapi) ---> top-k chunks +---> Graph Search (NetworkX Knowledge Graph) ---> top-k chunks | RRF Fusion: score = SUM(1 / (60 + rank_i)) | Top-K Chunks + Citations | LLM (Groq LLaMA-3 8B) --> Grounded Answer + [N] References
```

Why Hybrid Retrieval?

Retriever	Strength	Weakness
Semantic (ChromaDB+MiniLM)	Paraphrase matching, conceptual similarity	Misses exact law/formula names
BM25 keyword	Exact term match, law names, equations	No semantic understanding
Knowledge Graph	Entity relationships, multi-hop reasoning	Sparse, heuristic construction

Key Design Decisions

Component	Decision	Reason
Chunking	Section-aware + sentence boundary, 1500 chars / 200 overlap	Keeps semantically coherent content together
Embeddings	all-MiniLM-L6-v2 (384-dim)	Local, free; 95% quality vs. ada-002 at 10× lower cost
LLM	Groq LLaMA-3 8B	Free tier, 2× faster than GPT-3.5; sufficient for grounded Q&A;
Grounding	System prompt enforces context-only + [N] citations	Prevents hallucination on out-of-document questions

Results

~508	~200	~2–3 s	9
Chunks indexed	KG Nodes	API Latency	API Routes

Live Demo

```
cd rag_physics && uvicorn app.main:app --host 0.0.0.0 --port 8000 # Open: http://localhost:8000
```

Limitations & Next Steps

- **No reranking:** Add cross-encoder (ms-marco-MiniLM) after RRF for better precision.
- **Formula handling:** LaTeX extraction for structured equation parsing.
- **Multi-document:** Extend to full NCERT Physics Parts 1 & 2, or Chemistry.
- **Evaluation:** RAGAS framework with human-annotated QA pairs for faithfulness/relevance scoring.

Trade-off Discussion & Reproducibility

Model Selection Rationale

Decision	Chosen	Rejected	Reason
Grocery model	LightGBM	XGBoost, TFT	3× faster training; TFT overkill without GPU
ASL architecture	YOLOv8n	Faster R-CNN, EfficientDet	Single-stage → live FPS; nano → CPU deploy
RAG LLM	Groq LLaMA-3 8B	GPT-4	Free tier, 2× lower latency; GPT-4 not needed
RAG embeddings	MiniLM-L6-v2	OpenAI ada-002	Local, free, 95% quality at 10× lower cost

Validation Strategy Rationale

Use Case	Strategy	Why Not Random Split?
Grocery	Walk-forward 5-fold backtesting with 16-day gap	Time series: future leaks into training with random splits
ASL	Stratified 70/20/10 split	Classification: label imbalance; no temporal order
Physics RAG	Retrieval diagnostic endpoint + latency	No labelled test set; evaluated qualitatively

Pre-trained Artifacts

Artifact	Path	Size
LightGBM model	grocery_forecasting/models/lgbm_final.txt	22 MB
YOLOv8n weights	asl_detector/models/asl_yolov8_best.pt	6 MB
BM25 index	rag_physics/indexes/bm25_index.pkl	880 KB
ChromaDB index	rag_physics/indexes/chroma/	6 MB
Knowledge graph	rag_physics/indexes/knowledge_graph.gpickle	468 KB
Kaggle submission	grocery_forecasting/submissions/submission_lgbm.csv	736 KB

Quick-Start Commands

```
# UC1 Grocery Forecasting cd grocery_forecasting pip install -r requirements.txt jupyter
notebook notebooks/01_grocery_sales_forecasting.ipynb # UC2 ASL Detection (live webcam) cd
asl_detector pip install -r requirements.txt python demo.py --weights
models/asl_yolov8_best.pt # UC3 Physics RAG (web demo) cd rag_physics pip install -r
requirements.txt cp .env.example .env # add your Groq API key uvicorn app.main:app --host
0.0.0.0 --port 8000
```

External Resources Disclosed

- Kaggle ASL Alphabet dataset ([grassknoted/asl-alphabet](https://www.kaggle.com/grassknoted/asl-alphabet), CC BY-SA 4.0)

- Kaggle Store Sales competition dataset (Corporación Favorita)
- NCERT Class 12 Physics textbook (public domain)
- YOLOv8 pretrained weights — Ultralytics (AGPL-3.0)
- MediaPipe Hands — Google (Apache 2.0)
- all-MiniLM-L6-v2 — sentence-transformers (Apache 2.0)
- Groq LLaMA-3 8B API — Meta/Groq (Llama 3 Community License)
- AI-assisted code generation (GitHub Copilot) — disclosed, all logic understood and verified